HIMANSHU SRIVASTAV

Software Developer | Software Development Engineer (SDE) | Backend & AI Engineer

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EXECUTIVE SUMMARY

Software Developer / Software Development Engineer (SDE) with 2+ years of production experience building high-concurrency async backends (**FastAPI, Django, PostgreSQL 18, Redis, Celery**) and production deep learning / computer vision microservices (**RF-DETR, Detectron2, ResUNet, Geospatial AI**). Track record of shipping **8+ enterprise production applications** end-to-end, executing zero-downtime database migrations (**5.08M+ rows**), and architecting standardized **Deep Learning Bundles (DLBs)**. Experienced in fine-tuning on high-resolution aerial and orbital imagery (**USDA NAIP, Sentinel-1/2, ISRO LISS-4**), optimizing hardware throughput on **NVIDIA Ada Lovelace GPUs** (bfloat16 AMP), and building deterministic payload-driven inference services with automated GIS vectorization.

CORE TECHNICAL SKILLS MATRIX

Backend & Web Frameworks:	FastAPI, Django, Django REST Framework (DRF), Flask, adrf, Celery, Redis, SQLAlchemy 2.x async, aiomysql, WebSockets, Daphne / ASGI, Uvicorn, RESTful API Design, Microservices.
Databases & Systems:	PostgreSQL 18, MySQL, SQLite, Qdrant Vector DB, OpenSearch, Alembic, multi-tenant DB routing, N+1 query elimination, pgloader zero-downtime migrations (5.08M rows).
Computer Vision & DL:	RF-DETR (Lightweight DETR + DINOv2 ViT Backbone), Detectron2, Mask R-CNN, Faster R-CNN, FCN8, ResUNet-4B, Hungarian Bipartite Loss, Soft Dice Loss, Transfer Learning, Anchor Optimization, SAM, Grounding-DINO, DeepFace.
Geospatial & Remote Sensing:	Rasterio, Shapely, GeoPandas, GDAL, Multi-spectral/SAR (Sentinel-1 SAR, Sentinel-2 Optical, USDA NAIP 0.6m–1.0m, ISRO LISS-4 5.0m), GeoTIFF, Orthorectification, Cross-Tile NMS, Sliding-Window Tiling, Hanning-blend, NDVI, STAC.
Cloud, MLOps & Hardware:	AWS (EKS, S3, EFS), Docker containerization, Linux (WSL2 / Ubuntu), mixed-precision computing (bfloat16 / FP16 AMP), NVIDIA RTX 5000 Ada (Ada Lovelace), Snowflake / Fabric DB syncing, automated test suites (pytest).

WORK EXPERIENCE

Python Deep Learning Developer & Computer Vision Engineer

June 2026 – Present

GJ Maps Solutions Pvt. Ltd. | Udaipur, Rajasthan

Geospatial AI, Satellite Observation & Cadastral Mapping

- Built a continental agricultural field delineation pipeline with **Detectron2 Mask R-CNN (ResNet-50-FPN)** on 1.0m USDA NAIP aerial orthophotography, achieving **68.84% Mask AP■■** across 10,178 image chips and 4.8M vector annotations.
- Architected a 5.0m satellite cadastral parcel delineation engine for smallholder agricultural layouts across India (ISRO LISS-4 4-band rasters); custom-tuned Region Proposal Network (RPN) anchors ([[12], [24], [48], [96], [192]]) to capture smallholder plots (< 1 ha).
- Trained the India delineation model for **50,609 iterations (30 epochs)** using bfloat16 AMP on an NVIDIA RTX 5000 Ada (32GB VRAM), maintaining a steady ~3.8GB memory footprint and high training throughput (~3.2 batches/sec).
- Developed an end-to-end sliding-window inference pipeline (512×512 tiles, 64px overlap) with cross-tile spatial NMS deduplication, delineating **2,589 farm parcels in 15.5 seconds** on a full-scene GeoTIFF and exporting topologically valid vector polygons (GeoJSON) with dynamic acreage computation.
- Engineered a 4-band **ResUNet-4B** semantic boundary segmenter with Compound Boundary Loss (Boundary-weighted Cross-Entropy + Multiclass Soft Dice Loss), dynamic range normalization, and 2D Hanning-window blend reconstruction to eliminate tile seam artifacts.

Vision Transformers & Real-Time Object Detection (RF-DETR)

- Modernized privacy compliance and asset-tracking pipelines from legacy Faster R-CNN and YOLO models to **RF-DETR Medium (33.4M params, DINOv2 backbone)**, reducing inference latency while boosting localization accuracy.
- Built a unified multi-domain dataset of **37,830 images and 45,716 annotations** by aggregating Kaggle, Hugging Face, and 2,078 real-world street scenes from Goa, India; achieved **>97% mAP@50** and **72.1% strict mAP@50:95**.
- Fine-tuned RF-DETR for turbine blade defect identification (DTU Nordtank dataset), reaching **87.30% mAP@50** and an **85.77% F1 score** across surface pollution (*Dirt*) and structural fissures (*Damage*).
- Trained maritime vessel detection models on SAR satellite imagery (LS-SSDD), achieving **59.24% mAP@50** and **75.0% precision**; diagnosed training telemetry to halt execution at peak convergence (Epoch 13), preventing severe validation overfitting.
- Solved deep learning edge cases, including DETR zero-indexed foreground classification mappings, CPU-only worker fallback fixes, and spatial GSD mismatches between high-resolution sensors and orbital backbones.

Enterprise DL Platform Engineering & MLOps Infrastructure

- Designed and deployed standardized **Deep Learning Bundles (DLBs)** containing deterministic runtime specifications (metadata.json), catalog schemas (data.json), and isolated execution engines (inference/main.py).

- Eliminated silent runtime failures by replacing mock fallbacks with strict fail-loud exception contracts and non-blocking job progress reporting (update_job_status) compatible with distributed cloud worker pods.
- Built a dual-mode execution interface supporting platform task payloads (with EFS/POSIX path normalization) and standalone CLI debugging.
- Developed custom CLI packaging utilities (create_dlb.py) that bundle production weights while filtering out intermediate training checkpoints (.ckpt), caching layers, and database manifests.
- Coordinated database synchronization between primary catalog databases (Snowflake/Fabric) and regional inference queues (Pulse DB), managing automated artifact deployment to AWS S3.

Python Full Stack Developer
May 2025 – June 2026

MR Softwares | Ujjain, MP

- Designed, built, and shipped 8 production applications end-to-end (from database schema and API design to Docker containerization, Nginx SSL, CI/CD pipelines, and cloud deployment) serving enterprise clients independently.
- **eSchool Biometric & AI Attendance:** Led full framework migration from Django+DRF to FastAPI async with aiomysql + SQLAlchemy async connection pooling in 18 working days with zero downtime; integrated DeepFace FaceNet embeddings, sub-second Qdrant cosine matching, and Silent-Face Anti-Spoofing (MiniFASNetV2 PyTorch) liveness gate.
- **WaahApp (WhatsApp Business SaaS):** Led solo MySQL to PostgreSQL 18 + Azure to OCI migration of 5.08M rows with zero data loss in 2h 15m window; tuned PostgreSQL shared_buffers and indexes to slash API latency to sub-second responses.
- **Avvyx & LatLong:** Architected B2B SaaS PPF warranty system (FastAPI, SQLAlchemy 2.x, programmatic A4 PDFs) and real-time GPS tracking microservice (Django Channels, Traccar GPS, OpenSearch, geopy geodesic proximity alerts).
- **Hardware & Scraping:** Built multi-court scraping engine (6+ High Courts, Playwright/Selenium, Azure Functions) and PyQt6 desktop hotel management system with physical RFID card writers via pyserial.

Python Developer Intern | CodeAlpha (Remote: Oct 2023 – Jan 2024) • **Software Engineer Intern** | Oasis Infobyte (Apr 2023 – May 2023)

PROJECT DEEP DIVES

1. **Enterprise Deep Learning Bundle (DLB) Engine & Microservices Platform:** Engineered core DL runtime integration for XenTerra Pulse, replacing heterogeneous scripts with modular, container-ready bundles. Standardized runtime contracts across object detection, pixel classification, and instance segmentation. Solved platform hanging bugs via resilient try-except telemetry decoupled from inference loops; validated production packages with automated pytest suites.
2. **Multi-Continental Agricultural Field Delineation (Detectron2 Mask R-CNN & ResUNet):** Created instance & semantic boundary segmentation framework spanning USA (USDA NAIP 1.0m GSD; 68.84% AP■■ on 512×512 tiles with 64px overlap), Europe (Copernicus Sentinel-2 10.0m GSD), and India (ISRO LISS-4 5.0m GSD). Deployed 4-band ResUNet-4B with dynamic NDVI and Mask R-CNN with anchor tuning for smallholder farms; delivered full-scene georeferenced raster masks and WGS84 GeoJSON polygons.
3. **Edge & Aerial Surveillance via RF-DETR Vision Transformers:** Upgraded legacy CNN pipelines to Meta DINOv2-backed Detection Transformers across three critical domains: (a) *License Plate Anonymization:* Multi-threaded S3 ingestion, Grounding-DINO labels, joint training on 37.8k images (97.64% mAP@50), and Gaussian blurring; (b) *Windmill Defects:* Drone inspection data with Hungarian loss matching on RTX 5000 Ada (87.30% mAP@50); (c) *Maritime SAR Ship Detection:* Fine-tuned RF-DETR Medium on SAR GeoTIFFs (59.24% mAP@50, 75.0% precision, sub-2px box localization).
4. **Renewable Energy Mapping & Planetary Model Standardization:** Packaged research foundation models (Microsoft Global Renewables Watch FCN8 ResNet-50) into production DLBs with inlined inference, radiometric normalization, and spatial resolution scale calibration. Formulated dataset ingestion pipelines for road surface defects, urban asset tracking (Ashok Nagar & Udaipur), Microsoft Solar Farms, and NASA-IBM Prithvi foundation models (Burn Area, Flood, Crop ID).

KEY METRICS SUMMARY

Dataset Scale:	37,830 images / 45,716 annotations (License Plates); 10,178 chips / 4.8M vectors (USA Agriculture); 146k instances (India Agriculture); 2,995 drone frames (Windmill Defects).
Model Accuracy:	**License Plate:** 97.64% mAP@50, 72.13% strict mAP@50:95 | **Windmill Blade Defects:** 87.30% mAP@50, 85.77% F1 score **Agricultural Delineation (USA):** 68.84% Mask AP■■, 70.24% Box AP■■ | **Maritime Ship:** 59.24% mAP@50, 75.00% Precision
System Performance:	Full-scene satellite parcel delineation (Dharwad AOI 32) completed in **15.5 seconds (~20 FPS)** on RTX 5000 Ada, outputting **2,589 deduplicated parcels**.
Memory & Footprint:	Steady **~3.8GB VRAM** footprint during large-scale instance segmentation training using bfloat16 AMP. Average production .dlb bundle sizes maintained between **83 MB and 156 MB**.

EDUCATION & CERTIFICATIONS

B.Tech in Computer Science & Information Technology
IES IPS Academy, Indore, MP | 2021 – 2025 | GPA: 7.50 / 10

Python Developer Associate (PDAC-24) — ITechCert / AICERT (Jul 2024)
Computer Vision Builder — IIT Delhi (FITT / MP-IITD) (Mar 2024)